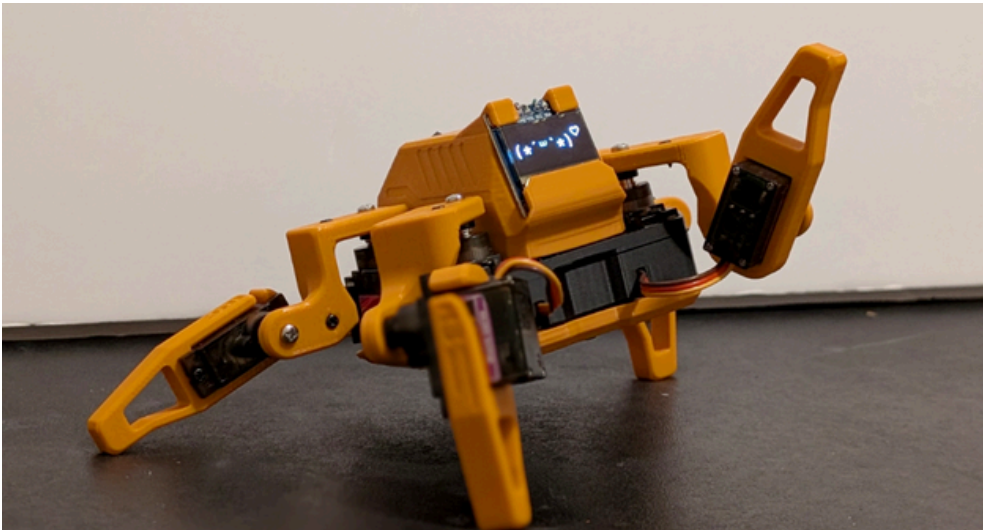




TA 212

END EVALUATION REPORT

QUADRUPED ROBOT



GROUP-8

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INTRODUCTION

This report documents the complete engineering process undertaken by Group8. It begins with the motivation and objectives of the project, followed by a detailed overview of the system's design, functional capabilities, and real-world applications. The project embodies the principles of mechatronics, combining mechanical design with electronic control to create a functional and scalable quadruped robot.

PROJECT OVERVIEW

The Quadruped Robot project is a specialized robotic platform built using the ESP32 microcontroller. The primary goal is to develop a stable, four-legged walking unit capable of navigating different environments. By integrating custom-dimensioned 3D-printed components, from the structural base plate to the dynamic leg links, the project demonstrates the successful synchronization of hardware and software.

Technical Design

The robot's architecture is centered around two main pillars:

Mechanical Kinematics: The robot utilizes a quadrupedal configuration with 8 Degrees of Freedom (DoF). Each leg is powered by two servo motors, allowing for precise gait cycles and stable locomotion.

System Control: The ESP32 acts as the brain of the robot, managing motor signals and wireless connectivity. This allows for real-time remote operation and pre-programmed movement sequences.

Potential Applications

The versatile design of the Sesame robot allows it to be utilized for various practical use cases:

Surveillance & Security: Its compact size and mobility make it ideal for monitoring restricted areas or navigating tight spaces where human access is difficult.

Search and Rescue: The multi-legged design offers superior stability on uneven terrain, providing a platform for inspecting hazardous environments or debris.

Remote Inspection: It can be used as a mobile unit to carry sensors for environmental monitoring, such as detecting gas leaks or temperature fluctuations in industrial settings.

The following sections provide a comprehensive overview of our design philosophy, component analysis, assembly process, and the collective learning outcomes achieved during this project.

ACKNOWLEDGEMENT

We would like to express our profound gratitude and sincere appreciation to everyone who supported and guided us throughout the duration of the Quadruped Robot Project.

First and foremost, we extend our heartfelt thanks to our instructor Mr. Mohit Law, Abhishek Kumar, for their invaluable mentorship, expert advice, and constant encouragement. Their insights and critical feedback were instrumental in navigating the challenges we faced during the design and assembly phases. We are also grateful to our institution for providing the necessary resources, 3D printer access, and a conducive environment that made this project possible. Our sincere thanks go to all our group members for their unwavering dedication, collaborative spirit, and hard work. The successful completion of this project is a testament to our teamwork and shared commitment.

Finally, we would like to thank our families and friends for their unwavering support and motivation throughout this endeavor.

ABSTRACT

This report details the design, development, and assembly of the Quadruped robot. The project aims to automate complex locomotion through a mechatronic system centered around the ESP32 microcontroller. By focusing on structural precision and hardware-software synchronization, this robot provides a scalable platform for exploring the principles of walking robotics and mechanical kinematics.

Technical Design & Fabrication: The robot's architecture is the result of a rigorous engineering workflow, focusing on precise modeling and efficient manufacturing:

- **CAD Modeling:** Every component, including the structural base plate and individual leg links, was meticulously designed and modeled in Autodesk Inventor. This ensured precise tolerances, proper alignment of the pivot points, and optimal housing for the electronic components.
- **Additive Manufacturing:** The mechanical assembly was fabricated using Creality 3D printing software. All parts were optimized for PLA printing to maintain high structural integrity while ensuring the robot remains lightweight for efficient actuation.
- **Kinematics & Actuation:** The quadruped configuration utilizes 8 high-torque servo motors (2 per leg) to achieve 8 Degrees of Freedom (DoF). This setup allows for the execution of precise gait cycles, including walking, waving, and resting animations.

The final prototype demonstrates the successful integration of CAD-based design with robotic automation. This project serves as a comprehensive engineering exercise, spanning from conceptual analysis in Autodesk Inventor to physical fabrication and testing.

MOTIVATION

Legged robots offer superior mobility in unstructured environments compared to wheeled systems, making them highly relevant for surveillance and exploration tasks.

This project focuses on developing a compact four-legged robot with limited actuation to study fundamental quadruped walking gaits while maintaining mechanical simplicity and low power consumption. Such a platform is well-suited for indoor surveillance, inspection of confined or uneven spaces, and rapid deployment scenarios. Additionally, the robot serves as an experimental testbed to compare gait efficiency, stability, and control strategies with more complex six-legged systems, contributing to a deeper understanding of legged locomotion applicable to real-world robotic applications.

TORQUE ANALYSIS

The bot has two links joint to two servos, say link1 and link2 of lengths approximately 1.5 inches and 2 inches (from CAD design). The worst case scenario is analyzed as such that all the weight of the bot is supported by one leg. The orientation of the servos are such that the link making the leg (link2) provides the required torque. The mass of the bot is 140 grams without breadboard (used for connections) and with breadboard is 270 grams. In order to safeguard a worst scenario the analysis is done using the latter mass i.e 270 grams.

$$\text{Torque (in kg-m)} = \text{Mass} \times \text{length}$$

$$\text{Torque (in N-m)} = \text{Force} \times \text{length}$$

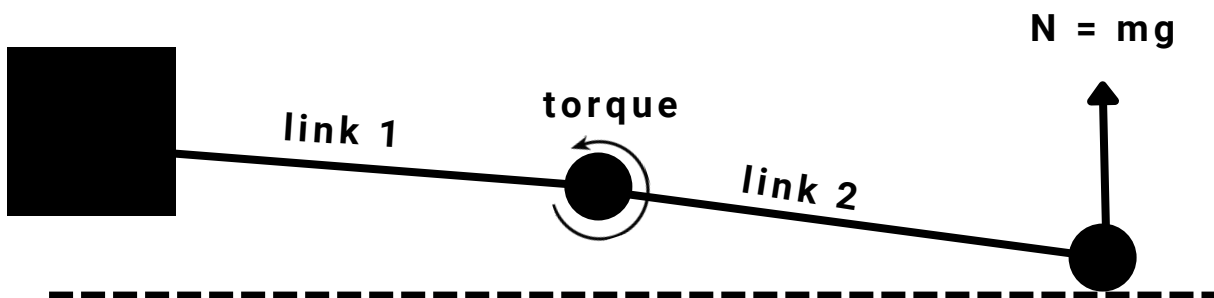
$$\text{Force} = mg = 0.270 \times 9.81 = 2.65 \text{ Newtons}$$

$$\text{Length} = \text{link2} = 2 \text{ inches} = 5.08 \text{ cm}$$

$$\text{Torque (in kg-cm)} = 0.27 \times 5.08$$

$$\text{Torque (in kg-cm)} = 1.3761 \text{ kg-cm}$$

The motors used for the following bot have stall torque 2 kg-cm at 4.8V and 2.5 kg-cm at 6.0V which is powerful enough to handle the worst case scenario even when Voltages from power source drops to as low as 4.8V at least.



POWER ANALYSIS - As mentioned above that the servos can deliver the required torque even at 4.8v thus we used an SMP and used to a buck converter to provide constant 5V and made a power rail to supply power to all 8 servos. Since the code gets executed line by line by a single micro controller there would not be issues of low powering (at a time a single servo draws most power).

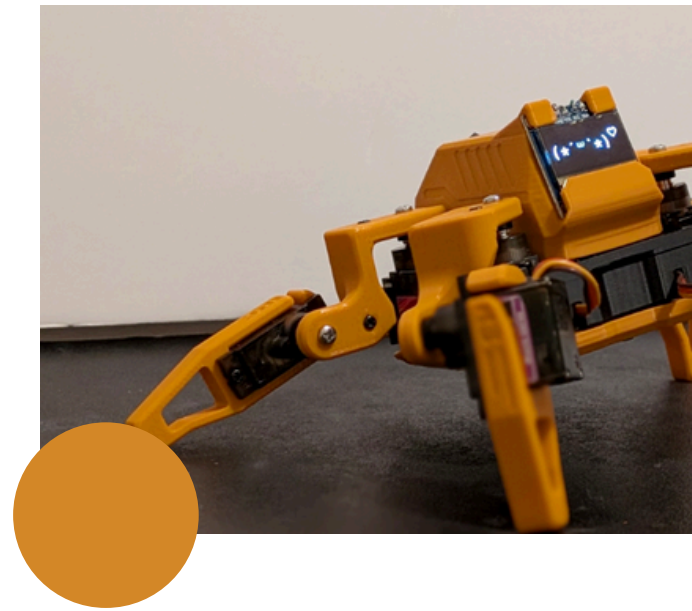
Software

Core Functionality and Code Structure

The software utilises the ESP32Servo library to interface with the hardware. The architecture is structured as follows:

- **Hardware Mapping:** Servos are assigned to specific pins (18, 19, 15, 2, 23, 22, 4, 5) using an *enum* for readable leg identification (R1-R4, L1-L4).
- **Initialization:** During the setup() phase, the code allocates four PWM timers and sets the servo frequency to 50Hz. It also defines pulse width limits (732 to 2929 microseconds) to ensure safe servo operation.
- **Motion Control:** The setServoAngle function serves as the primary movement engine. It incorporates a servoSubtrim array to correct mechanical misalignments and uses constrain() to prevent the motors from moving outside the 0–180° range.

Basic Poses and Movements



01 Standing (runStandPose)

Sets the legs to neutral angles (e.g., 45° , 135° , 180°) to support the robot's weight.

02 Pushups (runPushupPose)

Lowers the body by adjusting front legs (L1, R1) and then cycles the rear leg angles (L3, R3)

03 Turning (runTurnLeft/Right)

Rotates the robot by alternating "legset" movements in opposing directions.

04 Cuteness (runCutePose)

An expressive routine where the robot shifts its weight and tilts specific leg segments (R4, L4).

05 Waving (runWavePose)

Shifts balance to three legs while lifting and moving the front-right leg (L3) to mimic a wave.

06 Jump Pose (An Attempt)

We tried contracting and expanding the legs fast to attempt the jump pose, but it failed due to the servo motors' inability to respond fast enough and due to a lack of torque.

COST ANALYSIS

3D PRINTING

INTERNAL BASE: 3 HOURS

FOOT JOINT (4): $4 * 1 \text{ HOUR} = 4 \text{ HOURS}$

FEMUR JOINT (4): $4 * 1.5 \text{ HOUR} = 6 \text{ HOURS}$

TOTAL COST: $13 \text{ HOURS} * 300 \text{ Rs/Hour} = 3900 \text{ Rs.}$

RAW MATERIAL

ESP 32 = 300 Rs.

BATTERY:

SERVOS (8) = $8 * 100 = 800 \text{ Rs.}$

BUCK CONVERTER

TOTAL COST:

TOTAL COST: RAW MATERIAL + 3D PRINTING=